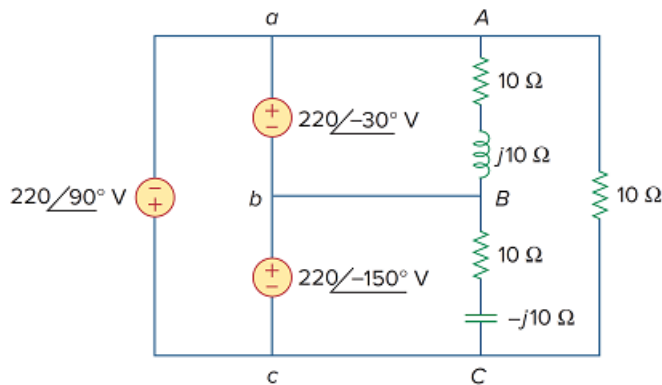


Problem

For the unbalanced circuit in Fig. 12.32, use *PSpice* to find the generator current I_{ca} , the line current I_{cC} , and the phase current I_{AB} .

Figure 12.32

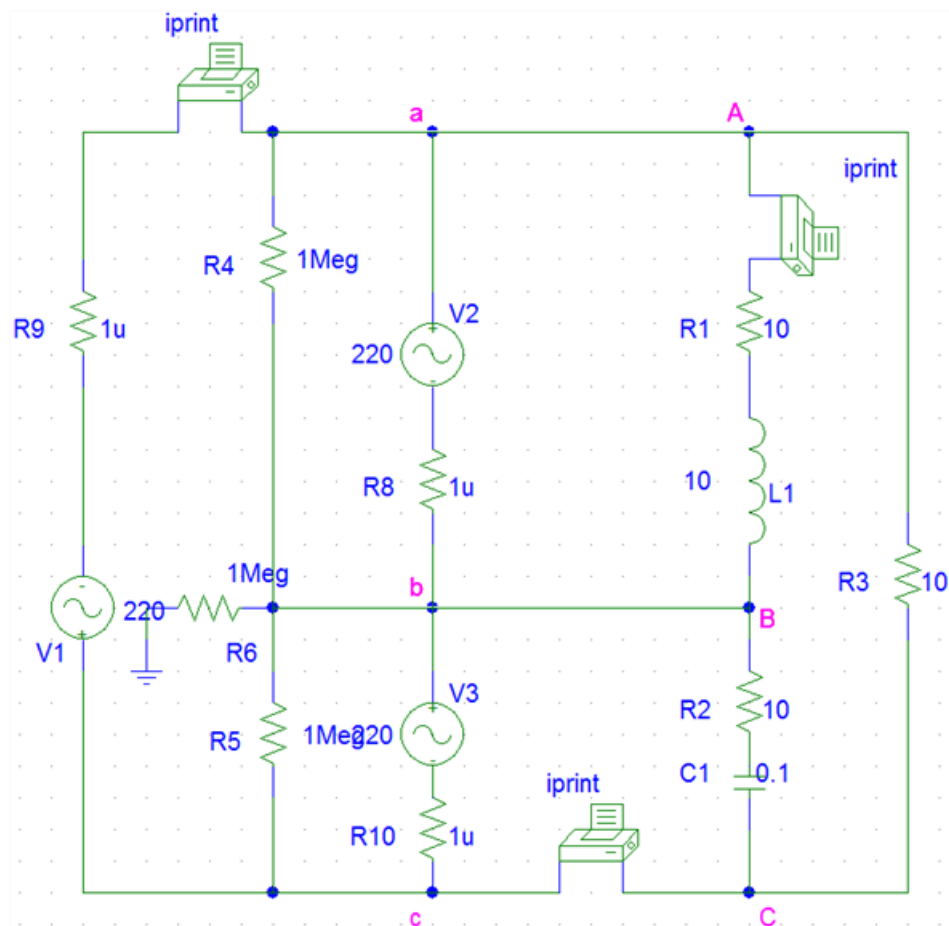


Step-by-step solution

Step 1 of 5

The schematic is shown in below:

Step 2 of 5



Step 3 of 5

Double click on 'VAC' voltage sources and set AC MAG and AC PHASE for given values.

As mentioned above, we avoid the loop of voltage sources by inserting a $1\text{ M}\Omega$ series resistor in the delta connected sources. To provide a ground node 0, we insert balanced wye – connected resistors ($1\text{ M}\Omega$ per phase) in the delta- connected source as shown in above schematic. Three Iprint pseudo components with their attributes are inserted to be able to get the required currents I_{ca} , I_{cc} and I_{AB}

Since the operating frequency is not given and the inductance and capacitances should be specified instead of impedance. We assume $\omega = 1\text{ rad/s}$ so that

$$f = \frac{1}{20}$$

$$f = 0.1592\text{ Hz}$$

Thus

$$L = \frac{X_L}{\omega}$$

$$L = \frac{10}{1}$$

$$L = 10\text{ H}$$

And

$$C = \frac{1}{\omega X_C}$$

$$C = \frac{1}{1 \times 10}$$

$$C = 0.1\text{ F}$$

We select analysis/setup/AC sweep and enter total pts = 1, start freq = 0.1592 and final freq = 0.1592. Once the schematic is saved, we select analysis/simulate to simulate the circuit. The output file includes.

Step 4 of 5

```
FREQ IM(V_PRINT7) IP(V_PRINT7)
```

```
1.592E-01 2.468E+01 -9.000E+01
```

```
FREQ IM(V_PRINT5) IP(V_PRINT5)
```

```
1.592E-01 3.725E+01 8.379E+01
```

```
FREQ IM(V_PRINT3) IP(V_PRINT3)
```

```
1.592E-01 1.555E+01 -7.501E+01
```

Step 5 of 5

Therefore

$$I_{Ca} = 24.68 \angle -90^\circ \text{ A}$$

$$I_{CC} = 37.25 \angle 83.79^\circ \text{ A and}$$

$$I_{AB} = 15.55 \angle -75.01^\circ \text{ A}$$